

Location, Time, and Observer Based Three-Valued Logic: A Minimal Formal System Where Truth is Never Absolute

Norbert Nopper

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Abstract

We present a minimal formal system in which the truth value of a proposition is never absolute, but is always evaluated with respect to three explicit parameters: a *location*, a *time*, and an *observer*. The system admits exactly three truth values—**T** (true), **F** (false), and **U** (unknown)—composed via Kleene’s strong three-valued connectives. Although three-valued logics, temporal logics, epistemic logics, and modal logics have all been studied extensively, the present contribution is not the discovery of new components but a *packaging* of known ideas into a single, minimal, self-contained primitive $V : P \times L \times \Theta \times O \rightarrow \{T, F, U\}$, in which location and observer are concrete and non-optional, and in which logical connectives are explicitly undefined across distinct (l, θ, o) frames. The result is a unified framework suitable as a starting point for systems that must reason about partial knowledge under spatial, temporal, and epistemic context.

Keywords: three-valued logic, temporal logic, epistemic logic, Kleene logic, contextual semantics.

1 Introduction

Classical two-valued logic forces every proposition into one of two states: *true* or *false*. In many practical settings—verification, distributed systems, knowledge representation, robotics, legal reasoning—this dichotomy is too coarse. Truth often depends on *where* an evaluation is performed, *when* it is performed, and *who* performs it.

We introduce a minimal logic that takes these dependencies as primitive. Every proposition is evaluated under an explicit tuple (l, θ, o) of location, time, and observer, and may take one of three values: true, false, or unknown. The system is intentionally austere: it contains no modal operators, no accessibility relations, and no transition systems. Its purpose is to serve as a compact, unambiguous core on which richer domain-specific layers (synchronization, trust, frame alignment) may later be built.

2 Truth Values

The system recognizes exactly three truth states, summarized in Table 1.

Table 1: The three truth values.

Value	Meaning
T (True)	Known to hold at this location, time, and for this observer.
F (False)	Known not to hold at this location, time, and for this observer.
U (Unknown)	Indeterminate at this location, time, and for this observer.

Classical logic forces every proposition into **T** or **F**. The present system acknowledges a third reality: that we do not always know.

3 Contextual Parameters

3.1 Location

Truth is a function of *where* it is evaluated. The same proposition at the same moment can hold different values in different locations:

$$\begin{aligned} V(p, l_1, \theta, o) &= T && \text{true in this location,} \\ V(p, l_2, \theta, o) &= F && \text{false in another,} \\ V(p, l_3, \theta, o) &= U && \text{unknown elsewhere.} \end{aligned}$$

“Location” is broadly defined: a physical place, a network, a jurisdiction, or a frame of reference. The essential point is that spatial and structural context matters.

3.2 Time

Truth is a function of *when* it is evaluated. The same proposition can yield different values at different moments—and this is not a contradiction, it is the system working as designed:

$$\begin{aligned} V(p, l, \theta_1, o) &= U && \text{not yet determined,} \\ V(p, l, \theta_2, o) &= T && \text{resolved to true,} \\ V(p, l, \theta_3, o) &= F && \text{later becomes false.} \end{aligned}$$

A future event is *unknown* before it occurs, *true* when it happens, and may become *false* if conditions change. Time is not optional: it is fundamental to every evaluation.

3.3 Observer

Truth is a function of *who* evaluates it. The same proposition at the same location and time can hold different values for different observers:

$$\begin{aligned} V(p, l, \theta, o_1) &= T && \text{true for this observer,} \\ V(p, l, \theta, o_2) &= F && \text{false for another,} \\ V(p, l, \theta, o_3) &= U && \text{unknown to a third.} \end{aligned}$$

“Observer” is broadly defined: a person, a sensor, a system, an institution, or any evaluating entity. Different observers may have different information, beliefs, or interpretive frameworks, and therefore arrive at different truth values for the same proposition at the same location and time. This reflects the epistemic reality that knowledge is observer-dependent.

4 The Valuation Function

The core primitive maps a proposition, evaluated at a location, time, and for an observer, to a truth value.

Definition 1 (Valuation). *Let P be a set of propositions, L a set of locations, Θ a set of time points, and O a set of observers. The valuation function is*

$$V : P \times L \times \Theta \times O \longrightarrow \{T, F, U\}.$$

For any $p \in P$, $l \in L$, $\theta \in \Theta$, and $o \in O$,

$$V(p, l, \theta, o) \in \{T, F, U\}.$$

Location, time, and observer are the *inputs*; the truth value is the *output*. No parameter is optional.

5 Logical Connectives

The three values compose through Kleene’s strong three-valued operations [2].

5.1 Conjunction

$\wedge$	T	U	F
T	T	U	F
U	U	U	F
F	F	F	F

5.2 Disjunction

$\vee$	T	U	F
T	T	T	T
U	T	U	U
F	T	U	F

5.3 Negation

p	$\neg p$
T	F
U	U
F	T

5.4 Implication

$\rightarrow$	T	U	F
T	T	U	F
U	T	U	U
F	T	T	T

5.5 Frame Discipline

All operations are evaluated at a given (l, θ, o) . When combining propositions, both must be evaluated within the same frame:

$$\begin{aligned} V(p_1, l, \theta, o) \wedge V(p_2, l, \theta, o) & \text{ valid: same } (l, \theta, o), \\ V(p_1, l_1, \theta_1, o_1) \wedge V(p_2, l_2, \theta_2, o_2) & \text{ undefined: cross-frame composition.} \end{aligned}$$

The logical connectives never escape the location, temporal, and observer frame.

Remark 1. Biconditional ($\leftrightarrow$) and quantifiers ($\forall, \exists$) are deliberately omitted to keep the system minimal. They can be defined in terms of the primitives above if needed.

Remark 2. Cross-frame composition—combining values from different (l, θ, o) tuples—is left undefined by design. Defining it would require explicit bridging rules (synchronization constraints, trust policies, observer-alignment or frame-alignment functions) that are domain-specific and beyond the scope of this core system.

6 Novelty and Related Work

The individual components of this system are well-established. Three-valued logics due to Łukasiewicz [1] and Kleene [2] define truth values without temporal or spatial indexing. Temporal logics such as LTL and CTL index truth by time but rest on classical two-valued semantics. Epistemic logics [4] index truth by agent knowledge but typically remain two-valued. Modal logics [5] index truth by possible worlds, again typically two-valued. Three-valued spatio-temporal logics have been explored in verification and robotics contexts.

The contribution of the present system is in its *formulation*, not in the discovery of its parts:

1. **Location as concrete and non-optional.** Not abstract “possible worlds” or “states”, but a first-class parameter representing physical places, networks, jurisdictions, or frames of reference.
2. **Observer as explicit and non-optional.** Truth is always relative to an evaluating entity, making the epistemic dimension a first-class parameter rather than an implicit assumption.
3. **The cross-frame constraint.** Logical connectives are explicitly undefined across different (l, θ, o) tuples, rather than silently mixing contexts.
4. **Minimalism.** No modal operators ($\Box, \Diamond$), no accessibility relations, no transition systems. Just $V : P \times L \times \Theta \times O \rightarrow \{T, F, U\}$ as a self-contained primitive.

This is a novel *packaging* of known ideas into a minimal, unified formal system—not a novel discovery.

7 Conclusion

We have presented a minimal three-valued logic in which every proposition is evaluated under an explicit triple of location, time, and observer. The resulting system is small enough to be specified on a single page, yet expressive enough to capture partial knowledge under spatial, temporal, and epistemic context. Cross-frame composition is intentionally left to future, domain-specific extensions.

References

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